Abu Henaf Rashid Ahmad Mufti

Applied AI Engineer & Research Enthusiast

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GitHub | LinkedIn | Portfolio

CAREER OBJECTIVE

Applied AI Engineer and research enthusiast building practical LLM applications, RAG systems, and machine learning solutions. Experienced in developing multi-modal pipelines, retrieval-augmented systems, and predictive models, with a focus on writing maintainable code and learning through hands-on development. Comfortable working across the stack, from data pipelines to deployed APIs, and continuously sharpening technical skills through applied projects and research.

EDUCATION

United International University — BSc in Computer Science and Engineering *Final-year Student*
Academia School, Mirpur, Dhaka — A Level & O Level, Edexcel Curriculum (Edexcel Award, Daily Star Award)

SKILLS

Languages: Python, SQL, C++, JavaScript
AI / ML: PyTorch, Scikit-learn, XGBoost, CNNs, Ensemble Learning, SHAP
LLM: LangChain, LangGraph, RAG, ChromaDB, Prompt Engineering
Backend: FastAPI, Streamlit, REST APIs
Tools: Git, Docker, Node.js, React

PUBLICATIONS

An Explainable and Scalable Framework for Compost Maturity Prediction (ACM Digital Library) *2024*
Peer-reviewed conference paper

– Developed a scalable framework integrating Ensemble Learning with Conversational AI to produce interpretable, explainable predictions for agricultural waste management.

RESEARCH

A Multimodal Risk Assessment and Monitoring Framework for Assisted Elderly Living *2026 – Present*
Final Year Design Project (FYDP)

– Building a signal-based ensemble classifier that fuses physiological and behavioural sensor data to flag health risks in assisted-living settings.
– Designing the alerting layer that surfaces detected risks to caregivers in real time.

DNA Protein-Site Binding Evaluation *Mid 2025 – Finalised*
Machine Learning Project

– Trained ML models to predict protein-site binding from biological sensorgram (BLI) data, converting high-dimensional signal curves into structured tabular features.

Pneumonia Lung Disease Classification *Mid 2025 – Finalised*
Digital Image Processing Project

– Built an image preprocessing and segmentation pipeline for chest X-rays, followed by classification using deep learning models optimised for limited GPU resources.

AQI Prediction and Generalisation *2026 – Present*
Peer Collaboration

– Trained and compared Random Forest, XGBoost, and CatBoost models on environmental data to predict air quality index, with SHAP used for feature-level interpretability.

PROJECTS

Unified MediChat — GitHub

- Designed a FastAPI-based AI health assistant with separate therapy, consultation, and vision workflows using LangGraph for request orchestration.
- Implemented a hybrid retrieval pipeline combining ChromaDB-based RAG, web search, and arXiv search to support evidence-informed responses.
- Built real-time streaming APIs using Server-Sent Events (SSE) with session memory, multimodal input support, and cloud deployment using Docker and Render.

Data Seeker — GitHub
- Developed a natural language to SQL pipeline with SQL validation and safety checks before query execution.
- Implemented automatic database schema inspection, heuristic SQL review, and visualization generation to improve query reliability and data exploration.
- Designed a two-stage LLM workflow separating SQL generation from analytical report generation for more structured outputs.

Research Assist — GitHub
- Developed a LangGraph-based research assistant that orchestrates paper retrieval, document reading, and literature synthesis within a stateful workflow.
- Automated the academic research process by integrating arXiv search, PDF processing, and publication-style report generation.
- Implemented streaming responses and tool-based agent workflows for multi-step research tasks.

Context ChatBot — GitHub
- Built a document question-answering system using PDF ingestion, semantic chunking, embeddings, and ChromaDB retrieval.
- Implemented a hybrid retrieval workflow that combines document-based RAG with live web search when uploaded documents lack sufficient context.
- Developed a Streamlit interface supporting document indexing, source-aware responses, and conversational interaction.

Dermatologist LLM Chat — GitHub
- Developed a multimodal dermatology assistant supporting voice input, skin image analysis, and spoken responses.
- Integrated speech-to-text, vision-language models, and text-to-speech into a unified diagnostic assistance workflow for educational use.
- Implemented alternative inference and speech synthesis backends to support different deployment configurations.

Web ScrapChat — GitHub
- Developed a full-stack AI search application using FastAPI, Next.js, and LangGraph for multi-step tool orchestration.
- Implemented real-time response streaming using Server-Sent Events (SSE) to improve interactive user experience.
- Designed a modular backend architecture integrating search workflows with conversational AI services.

COMPETITIONS

University Physics Competition (UPC) 2025
Accomplished Competitor — Ideation and Methodology